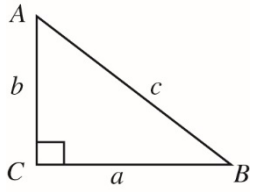
Solve a Right Triangle

Pythagorean theorem: $a^2 + b^2 = c^2$ $A + B + C = 180°$

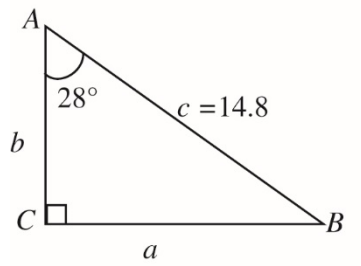


$$\sin\theta = \frac{\text{opp}}{\text{hyp}},\ \cos\theta = \frac{\text{adj}}{\text{hyp}},\ \tan\theta = \frac{\text{opp}}{\text{adj}}$$

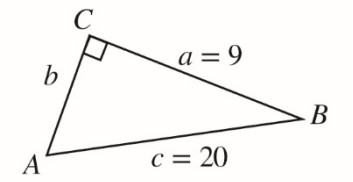
$$\csc\theta = \frac{\text{hyp}}{\text{opp}},\ \sec\theta = \frac{\text{hyp}}{\text{adj}},\ \cot\theta = \frac{\text{adj}}{\text{opp}}$$

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1. Solve the right triangle. Round the lengths of sides to 1 decimal place.



2. Solve the right triangle. Round the measurements to 1 decimal place.

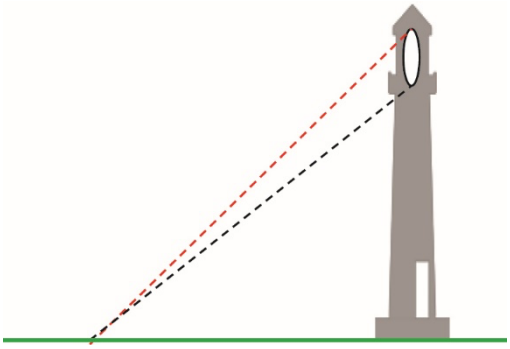


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Solve Applications of Right Triangles

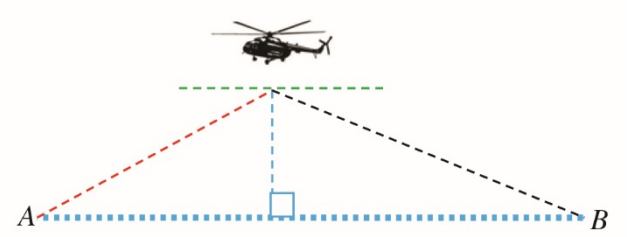
3. To determine the height of a window in the top of a lighthouse, two measurements are taken from the ground 90 feet from the base of a lighthouse. The angle of elevation from the ground to the bottom of the window is 36° and the angle of elevation from the ground to the top of the window is 42°.

   Find the height of the window to the nearest foot.



   From the given information, can we determine the height of the lighthouse?

4. An observer in a helicopter hovering 200 feet above the sea makes measurements to two buoys at points *A* and *B* below. The angles of depression are 44° and 35° respectively. Assuming the observer is in the same vertical plane as the buoys, what is the distance between the two buoys? Round to the nearest foot.



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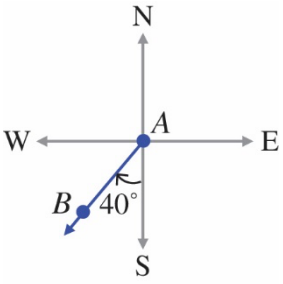
Compute the Bearing of an Object

The term *bearing* is used to convey the angular relationship from one point to another.

The **bearing** from point A to point B is the degree measure of the acute angle formed by ray $\overrightarrow{AB}$ and the north-south line.

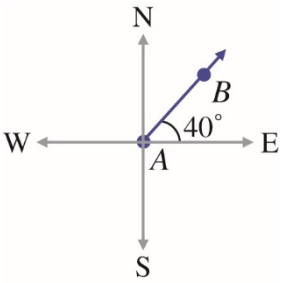
A bearing is denoted by the letter N or S (north or south), followed by the measure of the angle in degrees, followed by the letter E or W (east or west).

For example, a ship bearing S40°W is heading in a southwesterly direction. Specifically, the ship is traveling 40° to the west of due south.

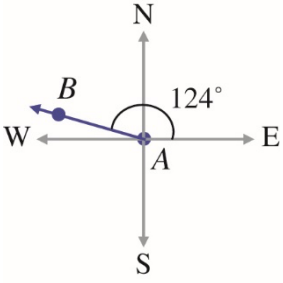


Find the bearing from A to B.

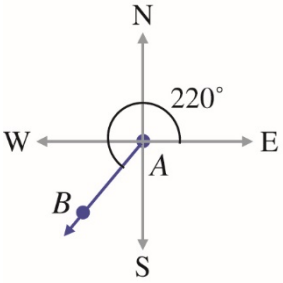
5.



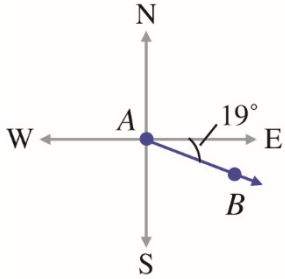
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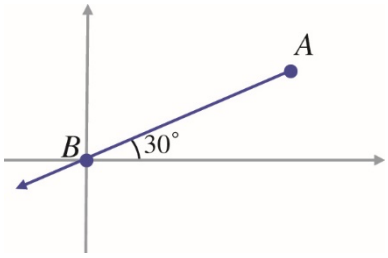
7.



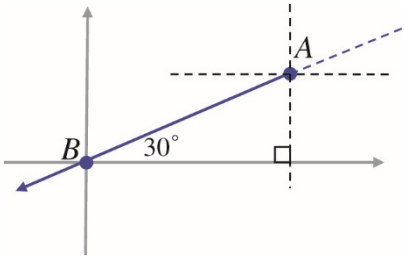
8.



9. Find the bearing from A to B.



Hint: Sketch a coordinate system with origin at point A.



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10. A boat leaves a marina and sails north for 30 minutes, then west for 2 hours. If the boat's average speed is 12 miles per hour, find the bearing for the return trip. Round to the nearest degree.

11. A plane leaves an airport at a bearing of N38°W and travels at 280 mph for 2 hours. The plane then makes a 90° left turn and travels 500 miles to its destination. Find the bearing required to locate the plane from the airport. Round to the nearest degree.

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